Interpreting Regression Results

MDragon Data Tools - Guide

Understanding Output

Slope (b_1)

Interpretation: Change in Y for one-unit increase in X

```
b₁ = 5.2
"Each additional X increases Y by 5.2 units"
```

Intercept (b_0)

Interpretation: Expected Y when X = 0

```
b₀ = 12.3
"When X=0, expected Y is 12.3"
```

⚠️ Only meaningful if X=0 is possible!

Correlation (r)

Value	Strength
0.0-0.3	Weak
0.3-0.7	Moderate
0.7-1.0	Strong

Sign:

- Positive (+): X ↑ then Y ↑
- Negative (-): X ↑ then Y ↓

R-Squared (R^2)

```
R² = 0.82
"82% of variation in Y is explained by X"
```

Guidelines:

- $R^2 < 0.3$: Poor fit
- $0.3 < R^2 < 0.7$: Moderate fit
- $R^2 > 0.7$: Good fit

Standard Error (SE)

Interpretation: Average prediction error

```
SE = 3.5
"Predictions typically off by ±3.5 units"
```

Common Mistakes

❌ Causation from Correlation

- Regression shows association, NOT causation
- Ice cream sales and drowning correlate (both increase in summer)
- Ice cream doesn't CAUSE drowning!

❌ Extrapolation

- Don't predict beyond data range
- Relationship may change outside observed values

❌ **Ignoring Outliers**

- One extreme point can drastically change line
- Always check scatter plot!

Reporting Results

Template

"We found a [strong/moderate/weak] [positive/negative] relationship between [X] and [Y] (r = __, R^2 = __). The regression equation is $\hat{y}$ = __ + __x. For each one-unit increase in [X], [Y] increases/decreases by __ units. The model explains __% of the variation in [Y]."

Example

"We found a strong positive relationship between study hours and exam scores ($r = 0.89$, $R^2 = 0.79$). The regression equation is $\hat{y} = 45 + 8x$. For each additional hour studied, exam scores increase by 8 points. The model explains 79% of the variation in exam scores."

Practice interpretation with real data in our calculators!
